EXECUTIVE BRIEFING · HEALTHCARE INTELLIGENCE

HIGHCARE: INTERACTIVE MEDICAL INTELLIGENCE



An end-to-end data analytics platform that unifies hospital operations, clinical caseloads, physician performance, and insurance claims into one operational truth.

POWER BI · PYTHON (PANDAS) · POWER QUERY · DAX · DATA MODELING

PRESENTED BY OMAR AHMED

AGENDA

WHAT WE WILL COVER

From the business case to the technology behind the platform, in eight sections.

1. Business Context & Challenges
2. Home Screen Architecture
3. Global Filters
4. Overview & Revenue
5. Hospitals & Utilization
6. Patients & Consultations
7. Doctors
8. Insurance Companies
9. Tech Stack & Recommendations
10. ARCHITECTURE & STRATEGIC ROADMAP

01 — THE OPERATIONAL BOTTLENECK

FRAGMENTED MEDICAL DATA WAS SLOWING EVERY DECISION

Critical healthcare telemetry was locked inside isolated operational and financial silos across facilities.



Isolated Systems

Hospital operations, patient flows, and financial claims lived in disconnected, static departmental reports.



Blindspot Handoffs

Leadership could not evaluate room occupancy alongside real-time revenue velocity or claim rejection rates.



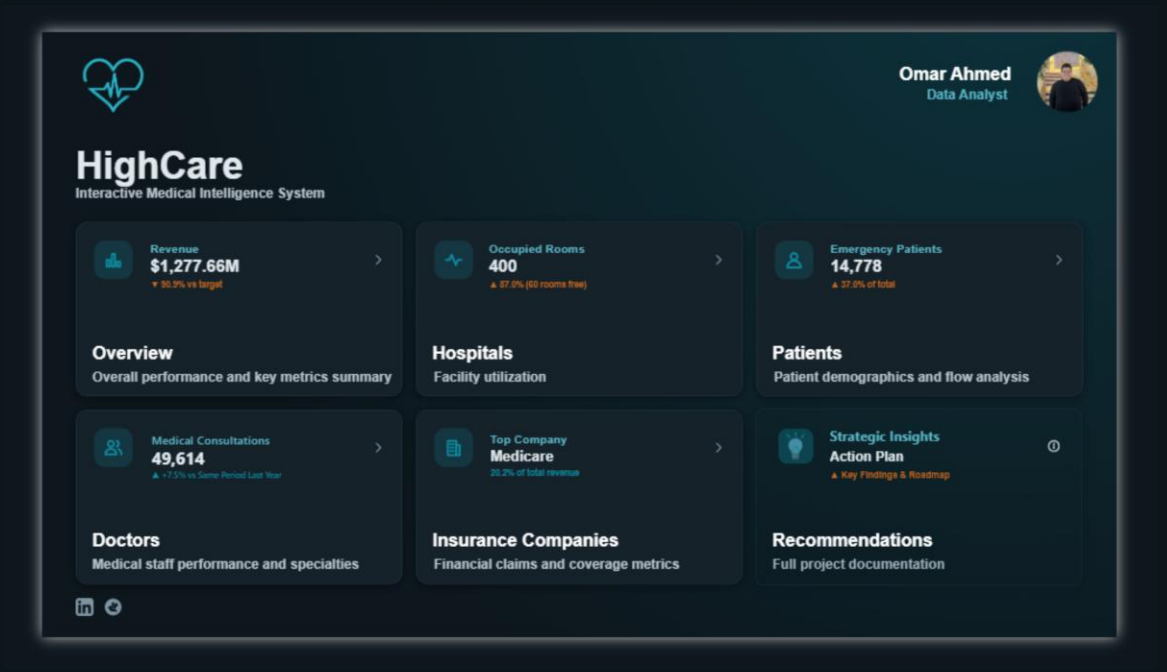
Unified Imperative

A single interactive BI cockpit was essential to convert scattered transactional records into actionable insights and recommendations.

02 — SYSTEM ARCHITECTURE

A CLEAN HOME SCREEN THAT GUIDES THE WHOLE SYSTEM

- **Overview & Revenue Tracking:** Real-time visibility into total revenues ($1,277.66M) and overall performance metrics across medical sectors.
- **Hospitals & Facility Utilization:** Instant monitoring of room occupancy, bed capacities, and operational efficiency.
- **Patient & Clinical Flow:** Detailed demographics tracking for emergency and regular patient influx.
- **Specialized Analytics:** Dedicated modules for medical consultations, top insurance companies (Medicare), and strategic project insights.



THREE GLOBAL FILTERS DRIVE THE ENTIRE SYSTEM

Medication, Year, and Quarter filters apply dynamically across all analytical pages.

GLOBAL FILTER 01

Medication

Filter by medication to explore patient cohorts and clinical outcomes.

GLOBAL FILTER 02

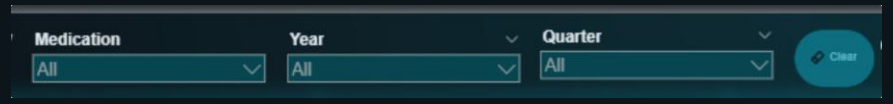
Year

Set the year to keep report views focused on the selected period.

GLOBAL FILTER 03

Quarter

Choose a quarter for a more focused view of report activity.



04 — FINANCIAL PERFORMANCE

REVENUE TRACKS AGAINST TARGET, LED BY KEY CONDITIONS

- **Total Revenue Generation:** HighCare tracking $1,277.66M overall revenue across medical operations.
- **Top Revenue Drivers by Medical Condition:** Diabetes ($215.07M), Arthritis ($215.05M), and Obesity ($213.78M) leading patient care financial returns.
- **Admission & Medication Balance:** Even distribution across Emergency ($418.34M), Urgent ($428.28M), and Elective ($431.04M) admission types.



Executive Finance Workspace · HighCare Analytics Engine

05 — CLINICAL CAPACITY

FACILITY UTILIZATION SHOWS WHERE CAPACITY IS TIGHT

- **Occupancy Saturation:** Room occupancy rates reveal network-wide bottlenecks before care degrades.
- **Key Capacity Ceiling:** 400 occupied rooms is the critical operational threshold for divert protocols.
- **Workforce Planning:** Dynamic utilization rates guide staffing, clinical nurse ratios, and on-call specialist scheduling.



Hospital Operations View · HighCare Resource Planner

06 — PATIENT DEMOGRAPHICS & RETENTION

DEEP-DIVE COHORT ANALYSIS & CLINICAL OUTCOMES

- **Core Patient KPIs & Diagnostics:** Tracks 39,948 total patients (evenly split 50% Male / 50% Female), monitoring a 14.8% returning patient rate alongside a 33.6% abnormal test result indicator.
- **Condition × Age Matrix:** Cross-references age cohorts (13 to 85 years) against medical conditions, highlighting high-concentration segments such as Arthritis in the 72–85 age group (1,813 cases).
- **Acquisition & Retention Velocity:** Evaluates quarterly patient flow through the "New vs. Returning Patients" longitudinal trend to measure long-term hospital retention.



Patient Analytics Module · HighCare Intelligence System

07 — CLINICAL WORKFORCE ANALYTICS

PHYSICIAN PERFORMANCE & CONSULTATION VELOCITY

- **Consultation Volume & Growth:** Tracks 49,614 total medical consultations (+7.5% vs. Same Period Last Year) to monitor clinical demand.
- **Caseload & Specialty Distribution:** Evaluates physician workloads, specialty coverage, and patient-to-doctor allocation across facilities.
- **Clinical Efficiency Metrics:** Identifies top-performing doctors and turnaround times to balance shift scheduling and prevent staff burnout.



Staff View · HighCare Intelligence Engine

08 — PAYER & CLAIMS INTELLIGENCE

INSURANCE COMPANIES CONCENTRATION & FINANCIAL COVERAGE

- **Medicare Revenue Share:** Highlights Medicare as the top insurance provider, driving 20.2% of total hospital revenue.
- **Payer Exposure & Risk:** Evaluates total financial claims, coverage ratios, and revenue distribution across public and commercial insurers.
- **Reimbursement Optimization:** Tracks claim cycles and payer reliance to guide contract negotiations and revenue diversification.



Payer View · HighCare Intelligence Engine

Powered by an optimized data model, advanced DAX measures, and custom HTML/CSS visual components.

TECHNICAL IMPLEMENTATION HIGHLIGHTS

Data Modeling & Architecture

Structured relational schema connecting hospital facilities, patient cohorts, clinical workforce, and insurance claims for instant cross-filtering.

Advanced DAX Engine

Custom time-intelligence and KPI measures tracking YoY consultation velocity (+7.5%), daily revenue targets ($1,661 vs. $1,910 target), and retention ratios.

Custom UI/UX Engineering

Built HTML/CSS visual cards, dynamic status pulse bars, zero-clutter navigation for a seamless SaaS-grade user experience , interactive cards and visuals.

DATA-DRIVEN STRATEGIC RECOMMENDATIONS

RECOMMENDATION 01

Optimize Facility Capacity & ER Flow

Route patient intake across secondary branches before hitting the 400-room capacity ceiling (87% occupancy), prioritizing the 14,778 emergency cases (37% of admissions).

RECOMMENDATION 02

Rebalance Payer Concentration

Expand commercial insurance partnerships to mitigate dependency on Medicare (20.2% of total revenue) and streamline claim reimbursement cycles.

RECOMMENDATION 03

Target High-Yield Chronic Care & Diagnostics

Focus clinical resources on top revenue-driving conditions — Diabetes ($215.07M) and Arthritis ($215.05M) — while investigating the 33.6% abnormal test result rate among returning cohorts (14.8%).

THANK YOU FOR EXPLORING HIGHCARE

Turning Complex Healthcare Telemetry into Actionable Executive Intelligence.



Omar Ahmed Ramadan Abdelaziz

Data Analyst & BI Developer | GIS Specialist

Power BI · DAX · SQL · Python · Excel · Tableau · Custom HTML/CSS UI

LinkedIn Profile

Professional network, project updates, and career history.

LINKEDIN

GitHub Portfolio

Full project repositories, code documentation, and analytical scripts.

GITHUB

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Interactive Report

Return to the live HighCare Power BI workspace.

POWER BI REPORT